CS 304 Software Engineering
Spring 2012 – FAST - NU Karachi Campus

Midterm II Part – A WAC Exam (April 5 & 6, 2012)
Course Instructors: Dr. Zubair A. Shaikh & Raza Noman

Max Time Allowed: 22 Hrs + 1 Hr **Max Marks: (100 + 50) = 150**

NOTE FOR Written Analysis of Case (WAC) EXAM:

1. This Part (WAC Exam) is of 22 Hrs and carries 75% Weightage towards Mid – II Grade. Mid – II has 15% weightage towards your Final Grade.
2. It's Handed out to ALL students at 11:00 AM PM (April 9th), its due back at exactly 9:00 AM (on April 10th) in the WAC Drop Box in HOD's Office. The Box shall be removed at exactly 9:05 AM.
3. All timings are Pakistan Standard Times. NO Exceptions / Emergencies, constraints in the extension can be granted.
4. The submission is in the shape of Handwritten on Answer sheet provided.
5. You are NOT Allowed to discuss with any other person on this case. You are responsible for the protection of your own work. Any student, whose work is found similar to other's work, will result in the award of "F" Grade in the course. All students shall submit the UNDERTAKING FORM with their submission.
6. For this exam, you must refer to the textbooks:
   a. Software Engineering by Roger S. Pressman (at least 6th Edition)
   b. Software Engineering by Ian Somerville (at least 6th Edition).
   c. Carnegie Mellon Institute's SEI site: http://www.sei.cmu.edu

PART "A" (Written Analysis of Case WAC exam).

Answer following questions in the sequence.

1) In your words write the summary of the whole project (approx 50 words / marks 5)

2) Requirements Elicitation and Analysis as defined by Sommerville (Sec 6.2 Chapter 6 in 6th Edition) mentions at least 6 different process activities for Requirements elicitation and analysis. These are: Domain Understanding, Requirements collection, Classification, Conflict Resolution, Prioritisation and Requirements Checking.

   3 component

   In your own words describe 3 different scenarios of the intended software systems (E-policing), translated into Requirements Elicitation and Analysis as mentioned by Sommerville. (10 Marks).

3) One of the major (perhaps the most important) activity in SE in Requirements Engineering and within that Requirements Analysis. Pressman in Chapter 7 Section 7.6 (6th Edition) describes building an Analysis Model. One of the techniques is to develop Analysis Patterns (section 7.6.2). They are like Design Patterns but meant for Requirements Analysis.

   In the Case Study Electronic FIR (Page 12 of the Case Study) is one of the Requirements of the system. Develop an Analysis Pattern for few of the activities of this Requirement. (10 Marks).

   use-case. and description

   semantic analysis pattern

4) E-Government Interoperability is a complex problem. Please refer to the technical report presented by Carnegie Mellon Software Engineering Institute in January 2012. See the link: http://www.sei.cmu.edu/reports/11tn014.pdf

   size, integer, secure

   US currency, no tax

   Data exchange, meaning each,

   In section 4.1 of the report it presents the Interoperability Goals. Describe some important Interoperability Goals in E-Police Case Study (10 Marks).

   process agreement

   software security

   action after the info exch.

   Why are these Goals Important. How will you implement them in E-Governance Implementation Model like in Figure 2 of the Report for E-Police Case Study. Describe all factors. (15 Marks).

   Legal, Political, socio cultural

   consistency with current policies

   accessibility for non-tech

   in that report and apply on E-police.

   what info is public

After you have developed the above model (also called Context Model as a generic term by Pressman, can you create the System Engineering Hierarchy of E-Police Study (section 6.2 Pressman). Please note that you don't have to create System Engineering Hierarchy for entire Case Study, but pick any one major customized applications mentioned in section 11.1.3 of the Case Report (15 Marks).

fit inter-operability model on use-case template.

5) What should be the ideal Waterfall/Spiral/Agile model with description of phases and activities. You must be able to identify the resource requirements and proper resource planning for the project. How will you do that, based on your knowledge of Project Planning, Scope and Software Models. (10 marks)

6) The Mission Risk Diagnosis is an approach presented by CMU SEI in technical report of February 2012 see full report, but specifically section 4. It is at: http://www.sei.cmu.edu/reports/12tn004.pdf

risks pg27&28.

Create set of Mission Risk diagnostic Drivers, like the ones mentioned in Table 2 Page 16 of the report for E-Police Case Study. Develop the Driver Value Criteria for a subset of E-Police Case Study (11.1.3.1.2 and 11.1.3.1.3). Assess the Mission Risk for these two Case Study components. (20 marks).

Elicit answer from Q-6

7) Do you see coherence in the pre-requirements presented in the E-Police. What do you think are key Software Engineering reasons that the project is still not implemented, despite the fact that well over 900 million Rupees have been invested, and the project was started in 2008 with a target completion by December 2010. (5 Marks).

GOOD LUCK and Remember the Undertaken.

Tomorrow you have One Hour Closed Book, Closed Notes Exam. You may be asked to draw a UML / interface design of any of the component and few basic questions, assessing your overall understanding of the theoretical knowledge and the Case.

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Midterm II Part – B Written Exam (April 10, 2012)
Course Instructors: Dr. Zubair A. Shaikh & Raza Noman

Max Time Allowed: 1 Hr **Max Marks: (25 in this part)**

NOTE
MOBILE Phones are NOT Allowed. If they are found on your desk, even if they are switched OFF, it will be considered as a Cheating Material.
Attempt ALL Questions. Be brief in answering.

Question 1 For any 2 of the following 3 Business Processes given in the case study, design appropriate UML or Business Process Diagram (10 marks)

BR. No.	1
Requirement Description	Daily Diary "ROZNAMCHA"
Role	SHO / Investigation Officer

BR. No.	2
Requirement Description	Character Certificate or Police Verification Certificate
Role	DSP / Senior Police Officer of the town.

BR. No.	3
Requirement Description	Computerization of challan / Ticket system
Role	Senior Traffic Police Officer.

Question 2 For any single Business requirement Mentioned above, design the possible User Interface, showing maximum functionalities (5 marks)

Question 3 How would you compare this (WAC + Written) exam based learning of Software Engineering concepts as compared to the conventional exam. List any Pros & Cons (5 marks)